

WiSe'25

Women in Computer Science

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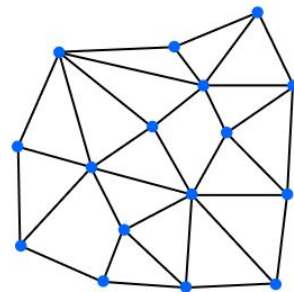
AUPE: Collaborative Byzantine fault-tolerant peer-sampling

Augusta MUKAM

December 15, 2025

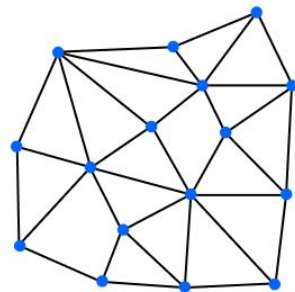
Distributed systems

- Collection of nodes running on separate virtual machines
- Nodes=computers
- Communicates through a network
- Provide services



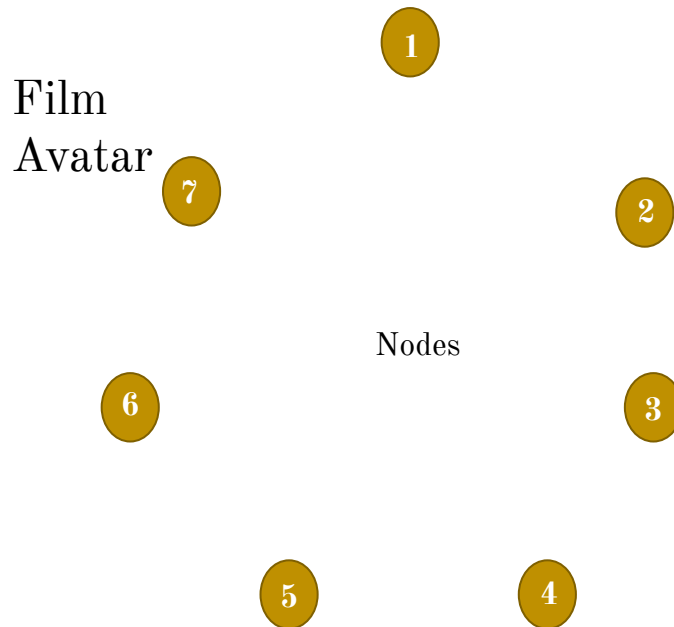
Distributed systems

- Collection of nodes running on separate virtual machines
- Nodes=computers
- Communicates through a network
- Provide services
- e.g., messaging application
- Different architectures



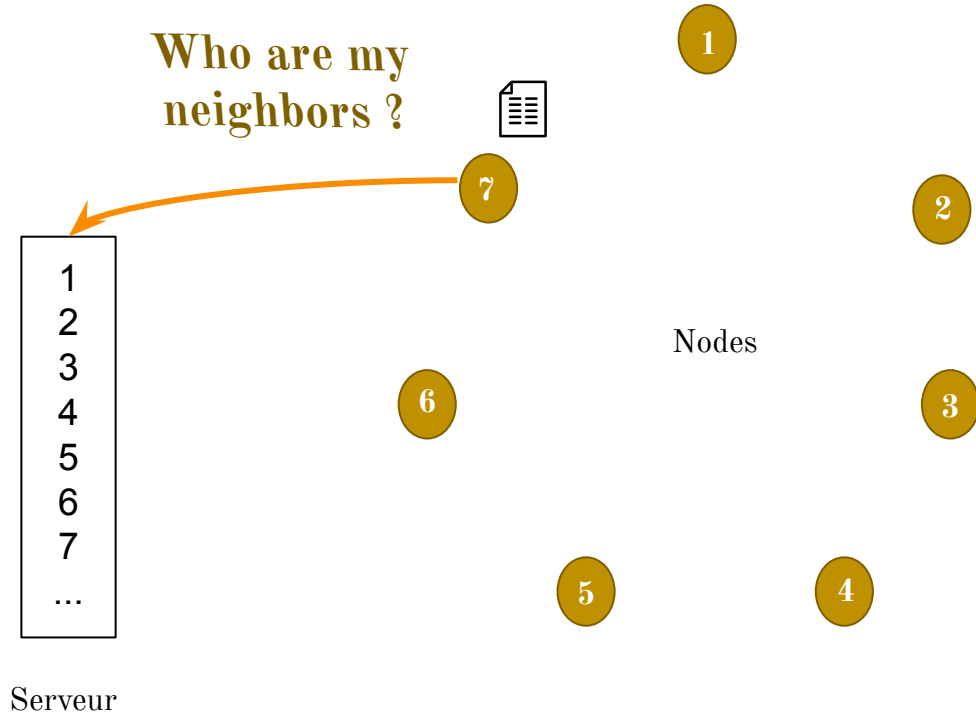
Distributed systems

- Nodes=computers
- Identifiers(ID), IP address



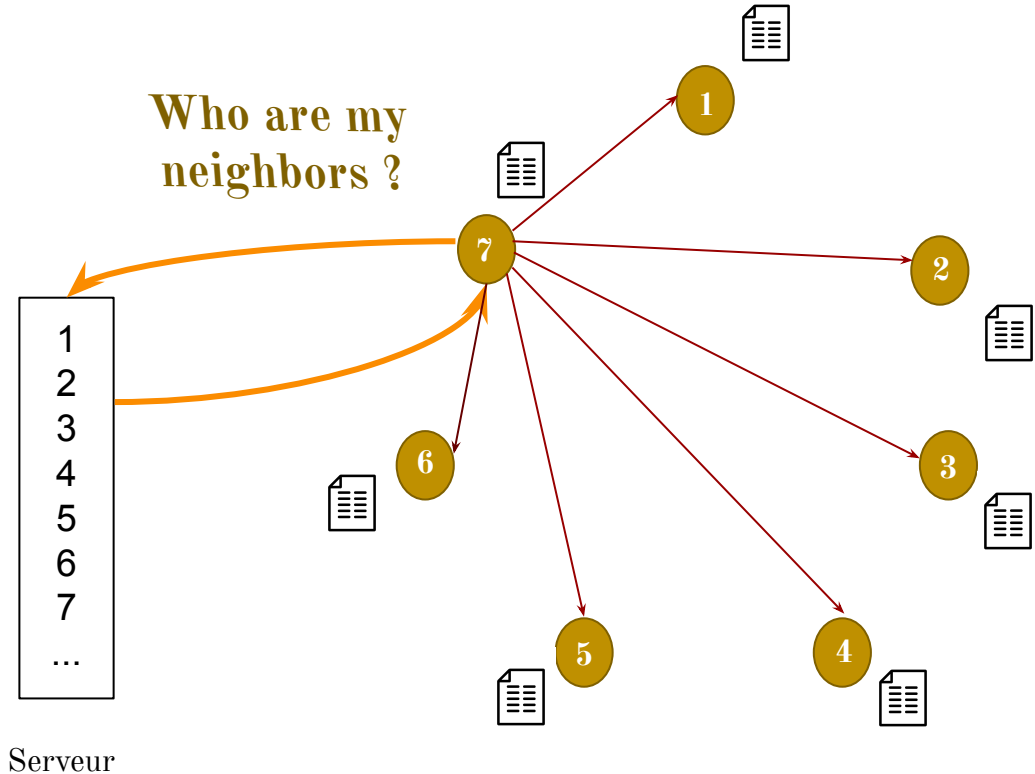
Distributed systems

- Broadcast message
- Centralized serveurur



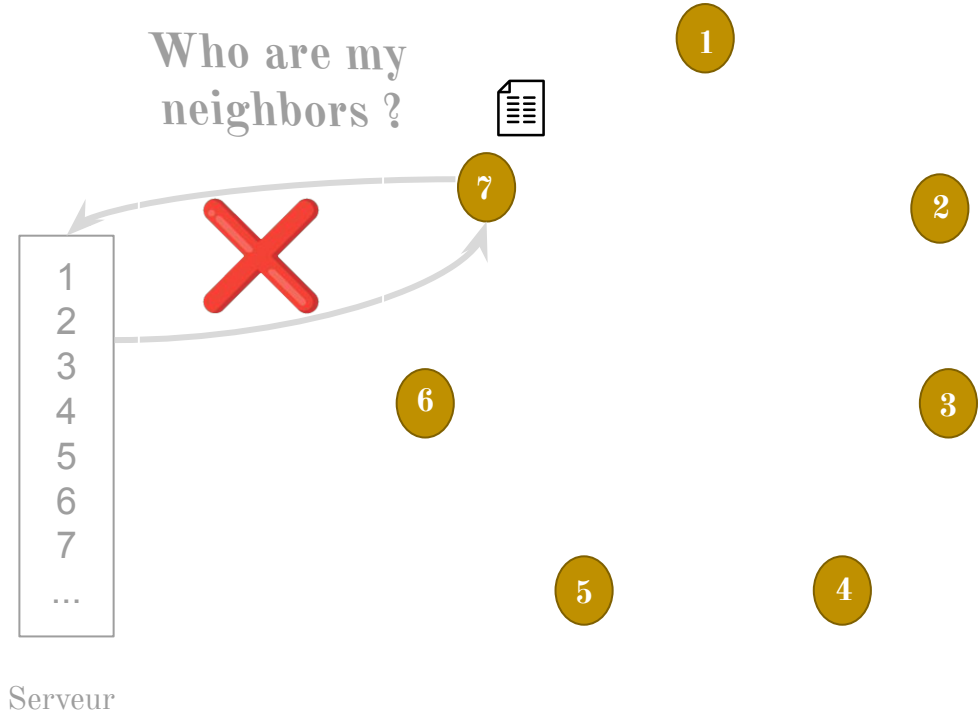
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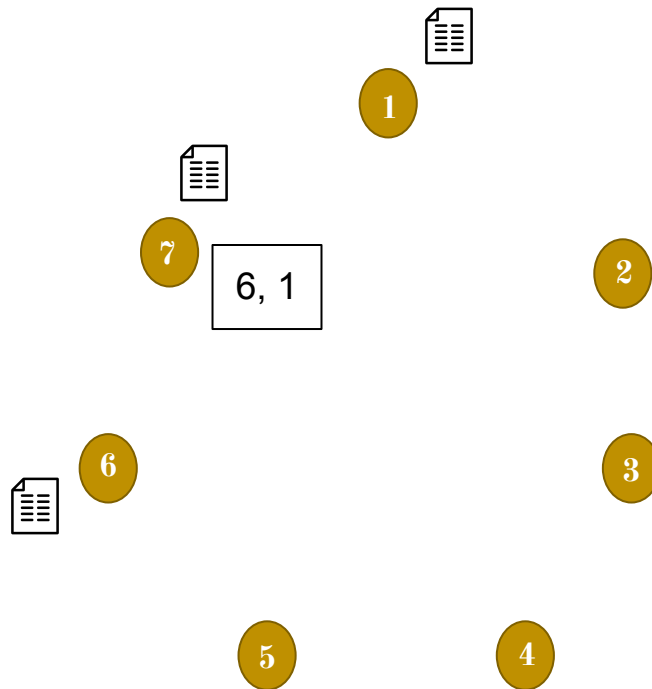
Distributed systems

- Large scale distributed systems
 - Many nodes
 - Continuous arrival and departure of nodes
 - Nodes failures
 - Too much synchronization costs !!!
 - Single point of failure



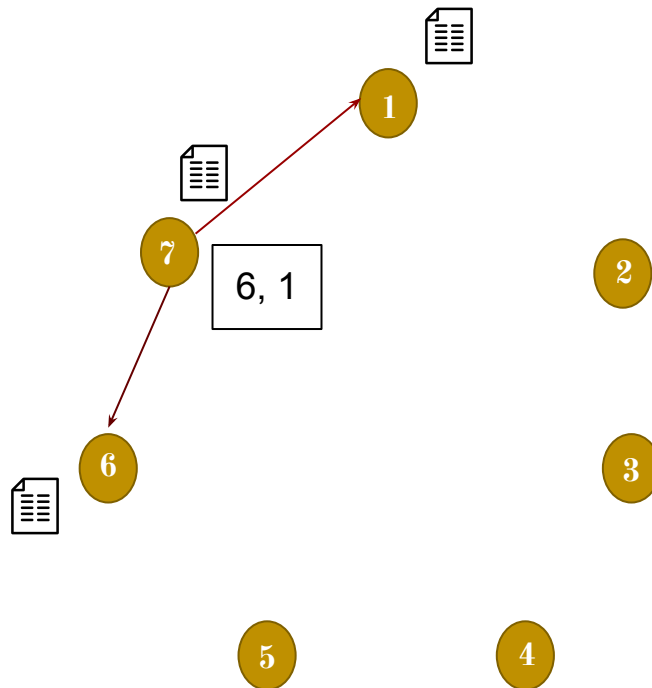
Distributed systems

- Large scale distributed systems
- Each node has a neighbor list



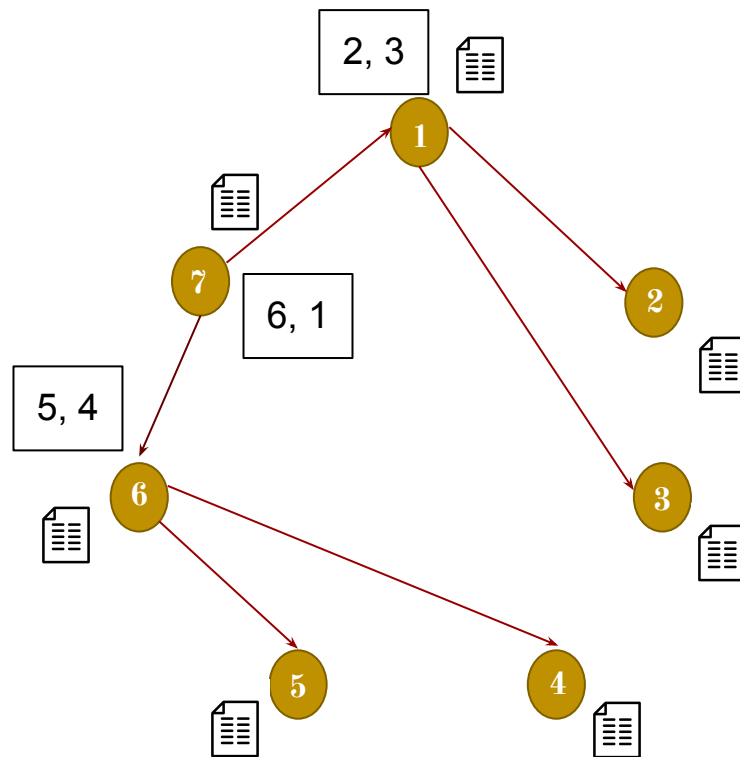
Distributed systems

- Large scale distributed systems
- Each node has a neighbor list
- Dissemination /Diffusion



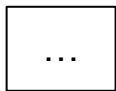
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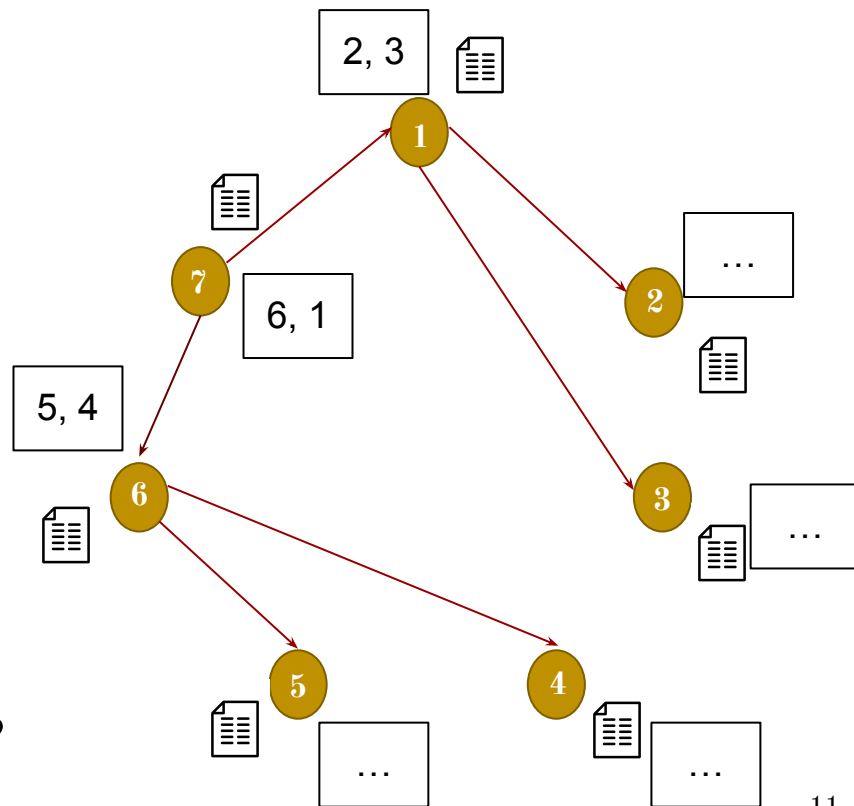


Distributed systems

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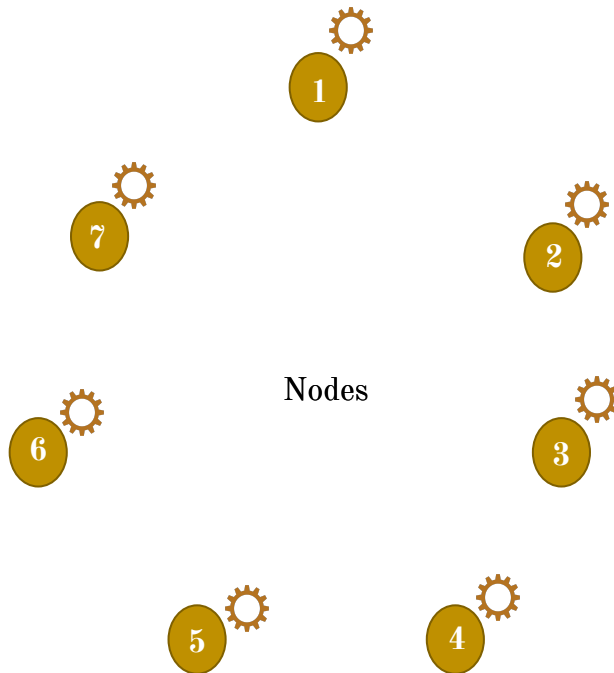


**How to build up-to-date
neighbor list for all nodes ?**



Distributed systems

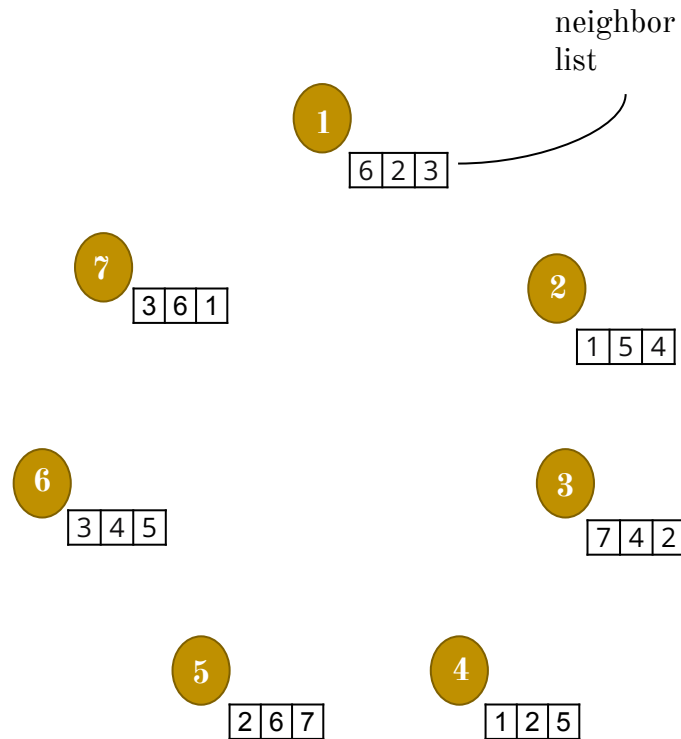
- **Peer sampling service**
 - Execute on each node
 - Maintain and update neighbor list
- Nodes exchange their membership information
- Goal: Have a distributed membership knowledge of the system



Peer sampling service

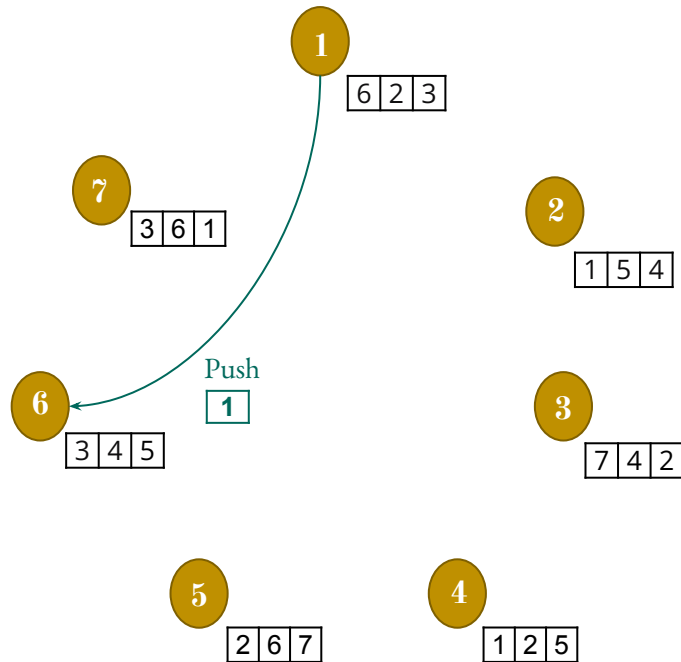
Peer sampling service

- Each node has a local **neighbor list**
- Small, fixed size



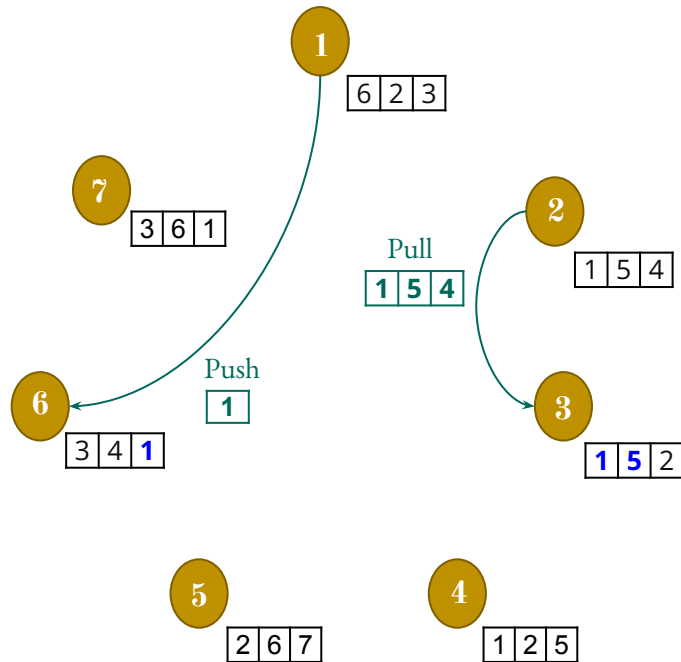
Peer sampling service

- Each node has a local **neighbor list**
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- Exchanges messages to mix their membership knowledge
 - **Push:** take my identifier



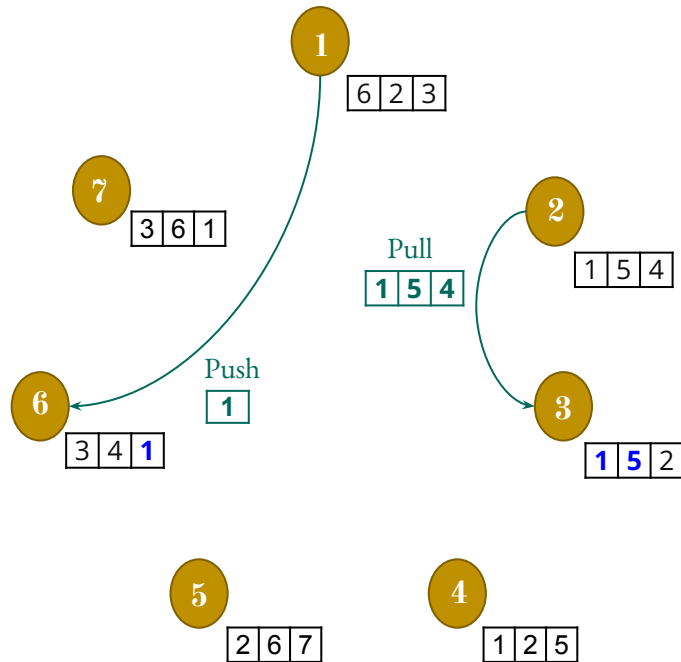
Peer sampling service

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 - **Pull:** Give me your neighbor list



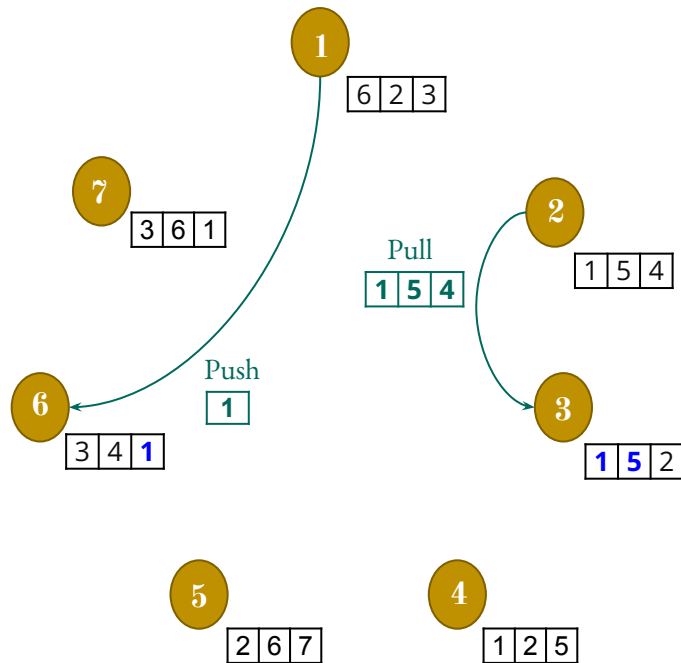
Peer sampling service

- Periodically (in rounds):
 - Exchange **Push** and **Pull** requests
 - Update view
 - Converge to a well-mixed state in few rounds



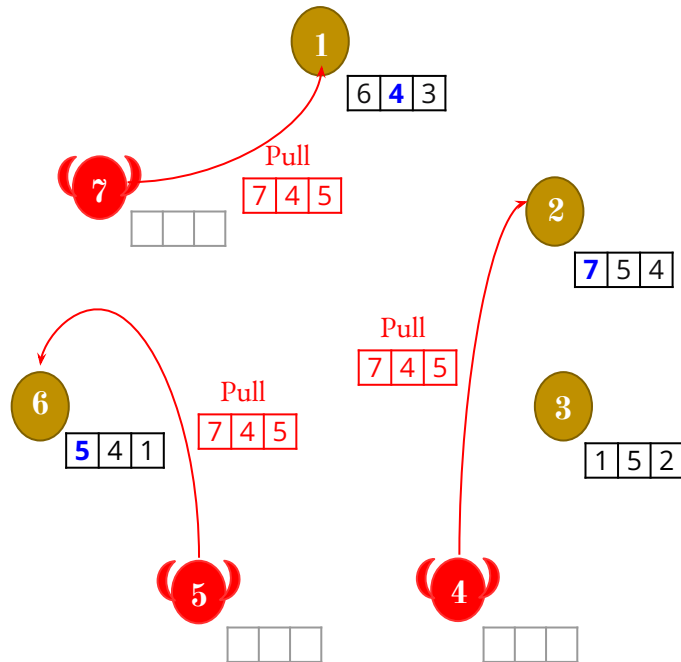
Objectives

- Ensure global network connectivity
 - Reach any node from any other node (avoid partitions)
- Manage node departure and arrival
 - $\frac{2}{3}$ of nodes leave after 1 hour
- Manage failures
 - Software, hardware, network



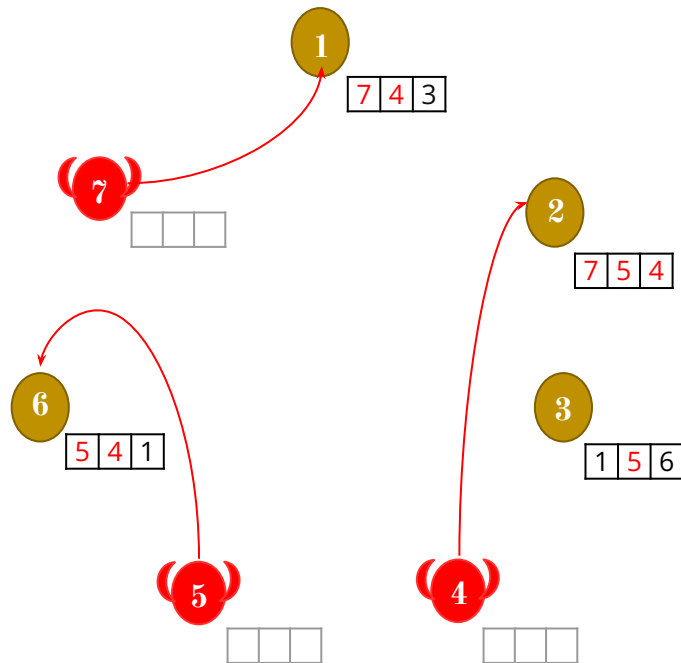
Problem

- Group of **malicious/Byzantine** nodes
- Promote nodes within their member group
- Messages: push, **false view reply**, ...



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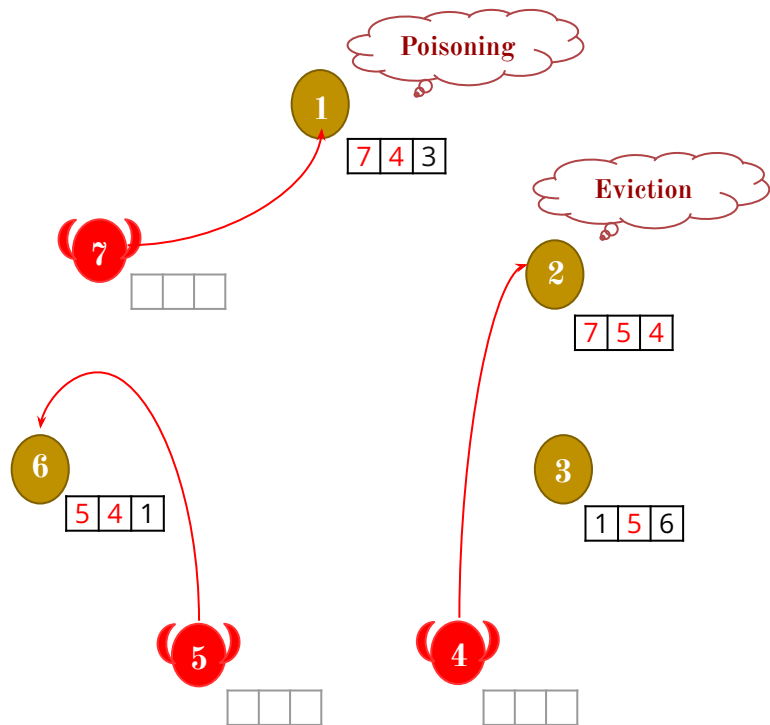


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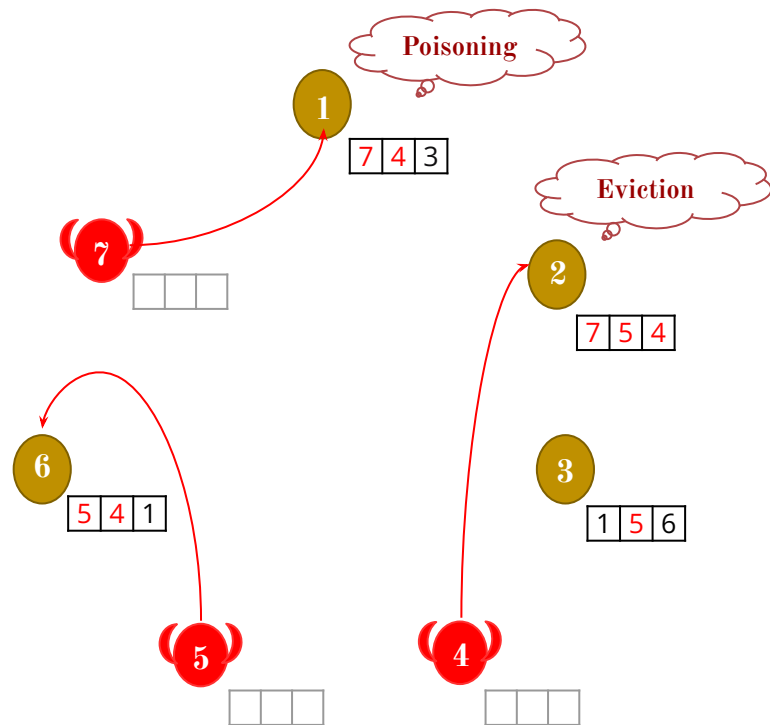
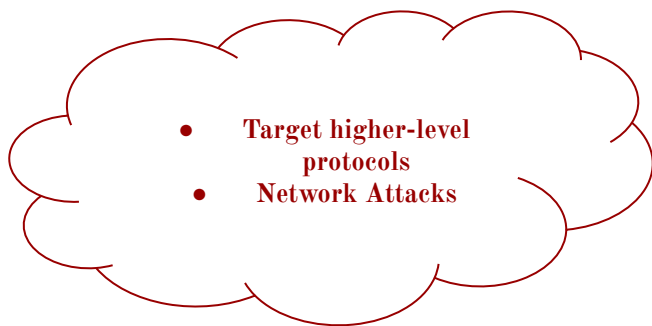
Eg: Bitcoin Eclipse attack

- Corrupted version of blockchain
- Manipulate their tokens



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Fault-detection

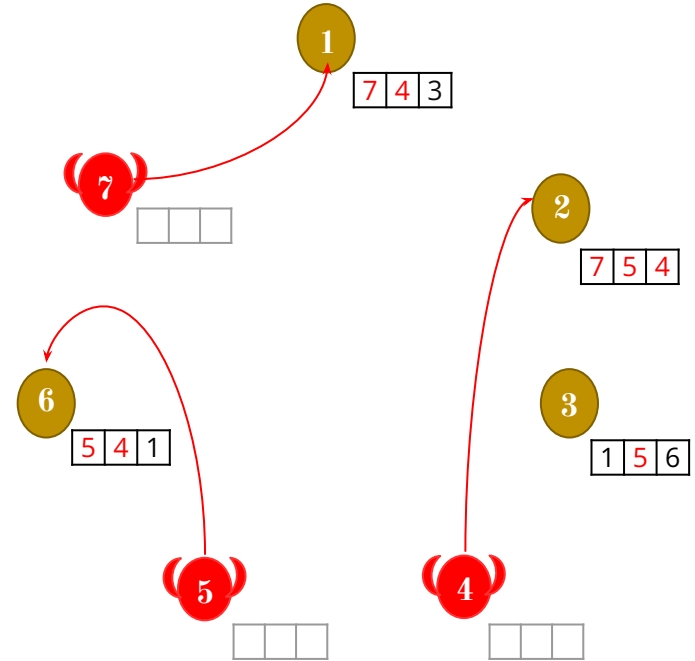
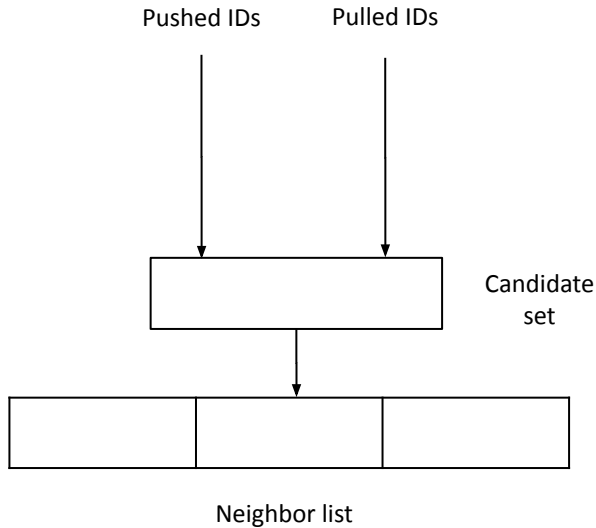
- Identify malicious nodes based on misbehavior proofs
- Punish faulty nodes
- Lead to major disruption

Fault-tolerance

- Tolerate malicious nodes
- Prevent them from polluting the system

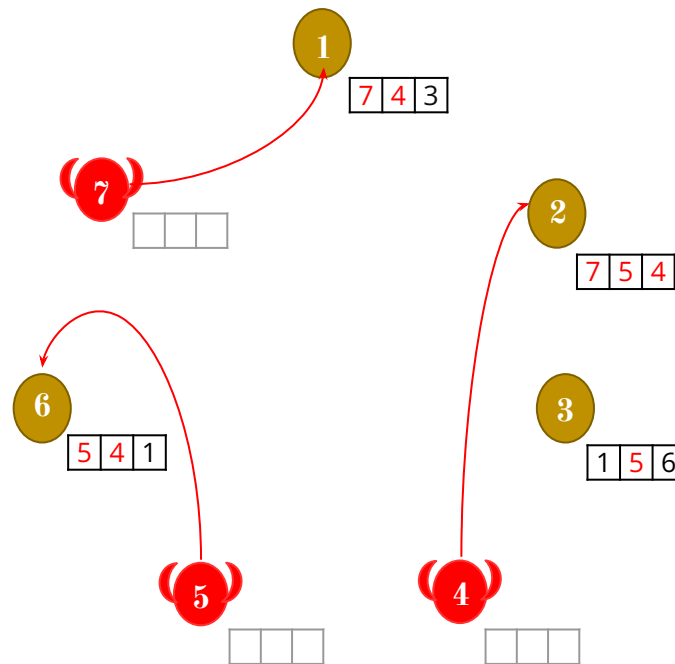
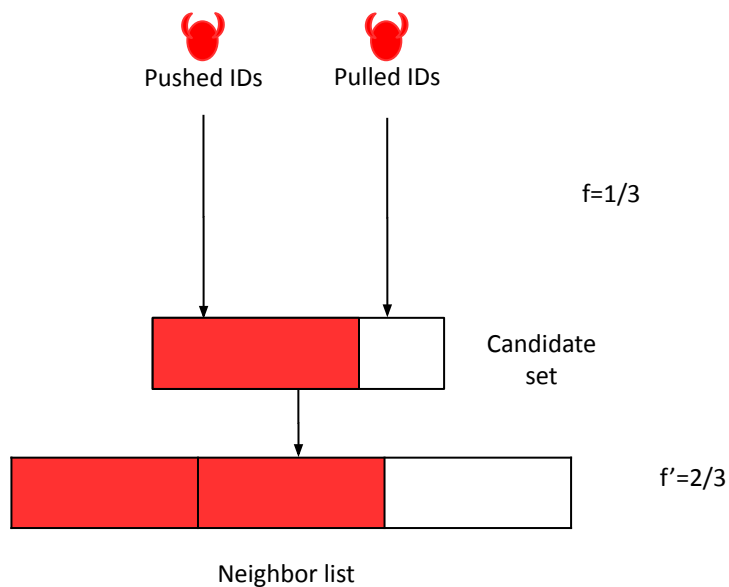
How Attack works?

- Naive approach
- Each round, perform an uniform sample



How Attack works?

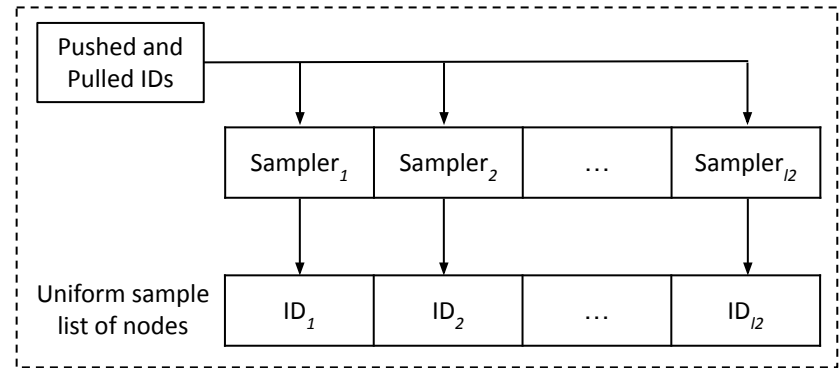
- Naive approach
- Each round, perform an uniform sample



Uniform sampling under adversarial context

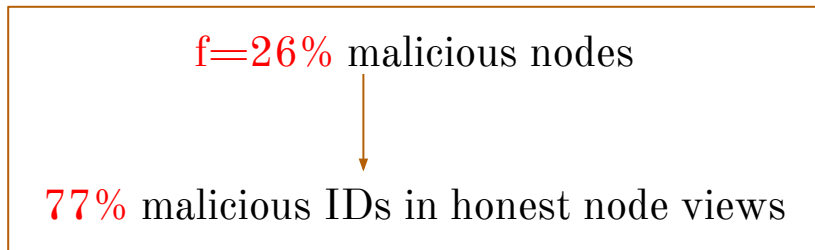
- **Brahms**
- Sampler: Min wise permutation hash functions
- Static views

Brahms

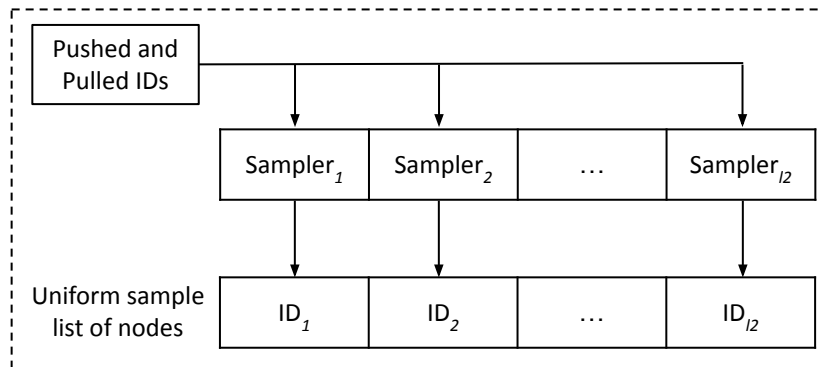


Uniform sampling under adversarial context

- **Brahms**
- Sampler: Min wise permutation hash functions
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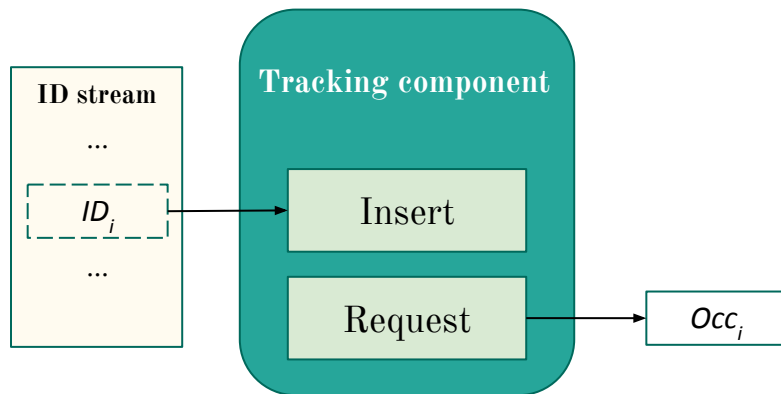
Brahms



AUPE Set Cleaner

Tracking component

- Record occurrences of received IDs in a tracking data-structure



Occurrence of node i (real or estimated): Occ_i

AUPE Set Cleaner

Debiasing component

- Transforms received stream to a more uniform one
- Probability of inserting into **sample memory**

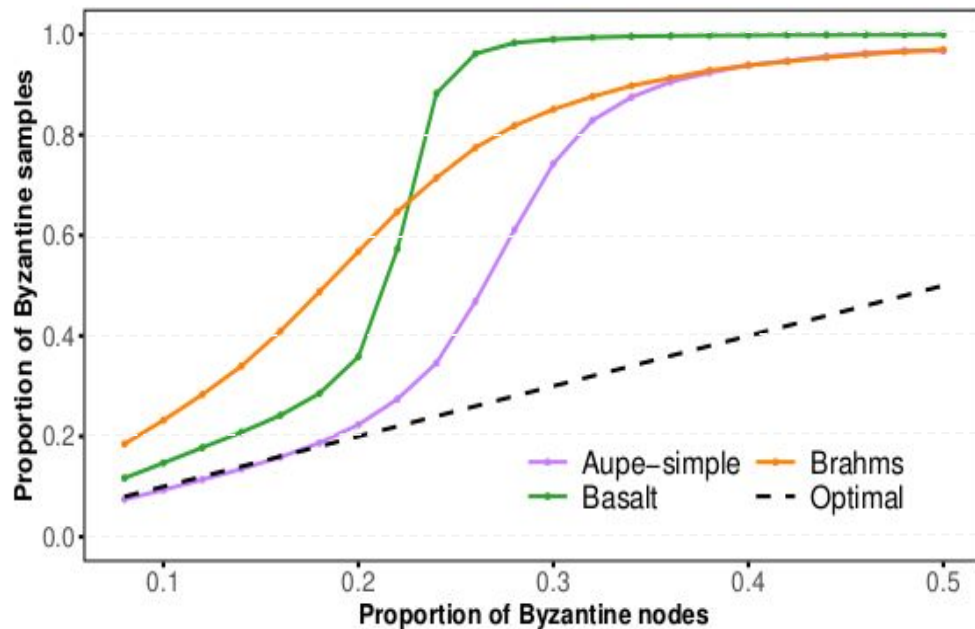
Probability of insertion of ID i : p_i

Minimum of all occurrences : *min*

$$p_i = \frac{\text{min}}{\text{Occ}_i}$$

System Tolerance improvement

Aupe-simple



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References

- *E. Bortnikov, M. Gurevich, I. Keidar, G. Kliot, and A. Shraer, “**Brahms**: Byzantine resilient random membership sampling,” in Proceedings of the Twenty-Seventh ACM Symposium on Principles of Distributed Computing, ser. **PODC ’08**. New York, NY, USA: Association for Computing Machinery, 2008, p. 145–154.*
- *E. Anceaume, Y. Busnel, and B. Sericola, “**Uniform node sampling service** robust against collusions of malicious nodes,” in 2013 43rd Annual IEEE/IFIP International Conference on Dependable Systems and Networks (DSN), 2013, pp. 1–12*
- *M. Jelasity, A. Montresor, and O. Babaoglu, “**Gossip-based aggregation in large dynamic networks**,” ACM Trans. Comput. Syst., vol. 23, no. 3, p. 219–252, aug 2005.*
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- *Augusta Mukam, Joachim Bruneau-Queyreix, and Laurent Réveillère. “**Aupe**: Collaborative byzantine fault-tolerant peer-sampling,” In 2024, 22nd International Symposium on Network Computing and Applications (NCA), pages 17–24, 2024.*